A Mean-Field Gradient-Flow Strategy for the Kaya Cup 2026

Mathematical Justification

Strategy design notes

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Abstract

We formalise the Kaya Cup as a controlled jump–diffusion on the unit disk and derive, from the exact game mechanics, a movement policy of the form $\mathbf{v} = \Pi_{\|\mathbf{v}\| \le v_{\max}}\big(-\nabla E(\mathbf{z})\big)$ — projected gradient flow on a physically motivated energy E. Each term of E is obtained from a governing mechanic: the fight coefficient from the win law, the danger and healing terms from Poisson event rates, the length scales from a heat-semigroup (diffusion) expansion, and the crowd term from a mean-field empirical measure. We prove the exact results (fight expectation, steepest-ascent optimality, Gaussian-convolution length inflation, Lyapunov descent) and cite the standard theorems used for the approximations (HJB verification, Feynman–Kac, LaSalle, laws of large numbers for empirical measures, mean-field games). We close with a proposition showing that *no* admissible policy attains win probability one, which delimits precisely what "optimality" can mean here.

Contents

1 The game as a controlled jump–diffusion

Definition 1.1 (State and dynamics). *Fix N players on the closed unit disk $\bar{D} = \{\mathbf{z} \in \mathbb{R}^2 : \|\mathbf{z}\| \le 1\}$. Player i has position $\mathbf{z}_i(t) \in \bar{D}$ and health $H_i(t) \in [0, 100]$. Writing $\mathbf{z} = \mathbf{z}_i$ for the focal player and $\mathbf{v} = \mathbf{v}_i$ for the velocity it chooses (the* control*, constrained to $\|\mathbf{v}\| \le v_{\max}$ with $v_{\max} = 1.2$), the position evolves as the reflected Itô diffusion*

$$d\mathbf{z}(t) \;=\; \mathbf{v}(t)\, dt \;+\; \sigma\, dW(t) \;+\; d\boldsymbol{\ell}(t), \qquad \mathbf{z}(0) \in \bar{D}, \tag{1}$$

where W is a standard planar Brownian motion, $\sigma > 0$ is the diffusion coefficient, and $\boldsymbol{\ell}$ is the boundary local-time term enforcing $\mathbf{z} \in \bar{D}$ [2]. Health H_i is a pure-jump process driven by independent Poisson clocks (Section 4). The time step is $\Delta t = 0.05$; (1) is its diffusive scaling limit, since the per-step increment $\mathcal{N}(0,1)\,\sigma\sqrt{\Delta t}$ has covariance $\sigma^2 \Delta t\, I_2$.

Definition 1.2 (Objective). *Let $T^* = \inf\{t : \#\{i : H_i(t) > 0\} \le 1\}$ be the end of the tournament. The focal player seeks a control maximising the win probability*

$$J^\star(\mathbf{z}, H, \cdots) \;=\; \sup_{\mathbf{v}(\cdot)} \mathbb{P}\big[\, i \text{ is the last survivor at } T^* \,\big]. \tag{2}$$

The randomness is genuinely irreducible: fights are Bernoulli, the NPC and combat clocks are Poisson, and $\sigma\, dW$ is uncontrolled. Section 10 makes the consequence precise. The remainder builds a tractable, mechanically faithful surrogate for (2).

2 Dynamic programming and its intractability

Theorem 2.1 (HJB verification for controlled diffusions [1, Ch. III]). *Let the full state be* $s = (\mathbf{z}_1, H_1, \ldots, \mathbf{z}_N, H_N)$ *with controlled generator* $\mathcal{L}^{\mathbf{v}}$. *If the value function* V *of* (2) *is sufficiently regular, it satisfies the Hamilton–Jacobi–Bellman equation*

$$0 \; = \; \sup_{\|\mathbf{v}\| \le v_{\max}} \Big\{ \mathbf{v} \cdot \nabla_{\mathbf{z}} V \; + \; \tfrac{\sigma^2}{2} \, \Delta_{\mathbf{z}} V \; + \; \mathcal{J} V \Big\}, \tag{3}$$

where $\frac{\sigma^2}{2}\Delta_{\mathbf{z}}$ *is the diffusion generator of* (1) *and* $\mathcal{J}$ *collects the jump (fight/NPC/recovery) terms. Conversely, a smooth solution of* (3) *with the correct terminal data equals* V, *and the maximiser* $\mathbf{v}^*(s) = v_{\max} \nabla_{\mathbf{z}} V / \|\nabla_{\mathbf{z}} V\|$ *is an optimal feedback control.*

Remark. *Two structural facts of* (3) *survive into the final policy regardless of the intractability below: (i) the control enters* only *through the linear term* $\mathbf{v} \cdot \nabla_{\mathbf{z}} V$, *so the optimal velocity is always aligned with a gradient and saturates the speed cap (Lemma 5.1); (ii) the second-order operator* $\frac{\sigma^2}{2}\Delta$ *is what couples the present decision to the diffusive future, and is the origin of the anticipation length scale of Section 6.*

The state s lives in $(\bar{D} \times [0, 100])^N$; for $N = 20$ this is an 80-dimensional space and (3) is subject to the curse of dimensionality [7]. We therefore replace V by a *potential surrogate* $-E$ built from one-step expected rewards, i.e. a myopic (greedy / one-step rollout) approximation [7, Ch. 6], and recover the strategic content lost by myopia through an explicit shaping term (Section 9). Sections 3–4 construct the exact one-step reward.

3 Exact fight microdynamics

Lemma 3.1 (Expected fight increment and the advantage coefficient)**.** *Suppose players i, j fight. Player i wins with probability $p_i = \dfrac{H_i}{H_i + H_j}$ and the loser forfeits 1 HP. Then*

$$\mathbb{E}[\Delta H_i \mid \textit{fight}] \;=\; -\frac{H_j}{H_i + H_j}, \qquad \mathbb{E}[\Delta H_i - \Delta H_j \mid \textit{fight}] \;=\; \frac{H_i - H_j}{H_i + H_j} \;=:\; s_{ij}. \tag{4}$$

Proof. $\Delta H_i = 0$ if i wins (probability p_i) and $\Delta H_i = -1$ if i loses (probability $p_j = \frac{H_j}{H_i+H_j}$), so $\mathbb{E}[\Delta H_i \mid \text{fight}] = -p_j = -\frac{H_j}{H_i+H_j}$. Symmetrically $\mathbb{E}[\Delta H_j \mid \text{fight}] = -p_i$, and subtracting gives $p_i - p_j = \frac{H_i-H_j}{H_i+H_j}$. □

Remark (Interpretation)**.** $s_{ij} \in (-1, 1)$ *is the* expected relative HP swing per exchange*. It is positive exactly when i is the healthier of the two: attrition under the win law is self-reinforcing (the leader loses HP more slowly than the trailer). This single quantity determines whether a neighbour is prey ($s_{ij} > 0$, attract) or predator ($s_{ij} < 0$, repel), and it is the coefficient of the pairwise force in* (6)*. Note $\mathbb{E}[\Delta H_i \mid \textit{fight}] \leq 0$ always: no fight is ever, in isolation, health-positive — the tension resolved in Section 9.*

4 Instantaneous expected health flux

Each interaction fires on an independent Poisson clock: player–player fights at rate λ while within $r_{\lim}$, Kaya at rate $\lambda_K = 2$, Butter at rate $\lambda_B = 1$ (only if $H < 50$), and boundary recovery at rate β (only if $H \le 25$ and $\|\mathbf{z}\| \ge 1 - r'$). The discrete probabilities $1 - e^{-\rho\Delta t}$ in the rules are exactly the event probabilities of a rate-ρ Poisson process on a step of length Δt [6].

Lemma 4.1 (Health drift)**.** *Conditional on the current configuration, the instantaneous drift of the focal player's health, $b_H(\mathbf{z}) := \lim_{\Delta t \downarrow 0} \Delta t^{-1}\mathbb{E}[\Delta H]$, is*

$$\begin{aligned} b_H(\mathbf{z}) \;=\; J(\mathbf{z}) := \;-\; \lambda \sum_{j \ne i} \mathbf{1}_{\{\|\mathbf{z}-\mathbf{z}_j\| < r_{\lim}\}} \frac{H_j}{H + H_j} \;-\; \lambda_K \,\mathbf{1}_{\{\|\mathbf{z}-\mathbf{z}_K\| < r_{\lim}\}} \\ +\; \lambda_B \,\mathbf{1}_{\{\|\mathbf{z}-\mathbf{z}_B\| < r_{\lim}\}} \mathbf{1}_{\{H < 50\}} \;+\; \beta\, \mathbf{1}_{\{H \le 25\}} \mathbf{1}_{\{\|\mathbf{z}\| \ge 1 - r'\}}. \end{aligned} \tag{5}$$

Proof. By the superposition theorem for independent Poisson processes [6], the pooled event stream is Poisson with rate equal to the sum of the active rates. Over $[0, \Delta t]$ the expected number of events of type k with rate ρ_k is $\rho_k \Delta t + o(\Delta t)$, and the probability of two or more events is $o(\Delta t)$. Multiplying each active rate by the expected HP increment of that event — $-H_j/(H + H_j)$ for a fight with j by Lemma 3.1, -1 for Kaya, $+1$ for Butter, $+1$ for recovery — summing, dividing by Δt and letting $\Delta t \downarrow 0$ yields (5). □

Remark (Smoothing)**.** *J is discontinuous through the indicators $\mathbf{1}_{\{\|\mathbf{z}-a\| < r_{\lim}\}}$. For a gradient policy we use mollified kernels $K_\ell(\mathbf{z} - a) = \exp(-\|\mathbf{z} - a\|^2 / 2\ell^2)$; Section 6 shows this is not an ad hoc convenience but the exact effect of averaging over the diffusive future, and it fixes ℓ.*

5 The steepest-ascent policy

Lemma 5.1 (Constrained steepest ascent)**.** *Let $g \in \mathbb{R}^2$, $g \neq 0$, and $c > 0$. Then*

$$\arg\max_{\|\mathbf{v}\|\le c} \langle g, \mathbf{v}\rangle \;=\; \frac{c\,g}{\|g\|}, \qquad \max_{\|\mathbf{v}\|\le c} \langle g, \mathbf{v}\rangle = c\|g\|,$$

and the maximiser is unique.

Proof. By the Cauchy–Schwarz inequality $\langle g, \mathbf{v}\rangle \le \|g\|\,\|\mathbf{v}\| \le c\|g\|$. Equality in Cauchy–Schwarz holds iff $\mathbf{v} = \alpha g$ for some $\alpha \ge 0$, and the cap forces $\|\mathbf{v}\| = c$, i.e. $\alpha = c/\|g\|$. Uniqueness is the strictness of Cauchy–Schwarz off the ray $\mathbb{R}_{\ge 0}\, g$. □

Definition 5.2 (Energy and policy)**.** *Define the energy*

$$E(\mathbf{z}) \;=\; \underbrace{-\,J(\mathbf{z})}_{\textit{expected loss}} \;+\; \underbrace{E_{\text{crowd}}(\mathbf{z})}_{\textit{mean field, §7}} \;+\; \underbrace{E_{\text{home}}(\mathbf{z})}_{\textit{radial camp, §8}} \;-\; \underbrace{g_{\text{end}}\,\Phi_{\text{prey}}(\mathbf{z})}_{\textit{shaping, §9}}, \tag{6}$$

with J mollified as above, and set the feedback control

$$\mathbf{v}(\mathbf{z}) \;=\; \Pi_{\|\mathbf{v}\|\le v_{\max}}\big(-\nabla E(\mathbf{z})\big) \;=\; \begin{cases} -\,v_{\max}\,\dfrac{\nabla E(\mathbf{z})}{\|\nabla E(\mathbf{z})\|}, & \nabla E(\mathbf{z}) \neq 0, \\ 0, & \textit{else.} \end{cases} \tag{7}$$

Proposition 5.3 (The policy is one-step optimal for the health surrogate)**.** *Among all admissible velocities, (7) with $E = -J$ maximises the instantaneous expected rate of change of J along the induced motion, $\frac{d}{dt}J(\mathbf{z}(t))\big|_{drift} = \langle \nabla J, \mathbf{v}\rangle$.*

Proof. By the chain rule the drift contribution of the control to $J(\mathbf{z}(t))$ is $\langle \nabla J(\mathbf{z}), \mathbf{v}\rangle$. Apply Lemma 5.1 with $g = \nabla J = -\nabla E$ and $c = v_{\max}$: the maximiser is $v_{\max}\nabla J/\|\nabla J\| = -v_{\max}\nabla E/\|\nabla E\|$, which is (7). □

Remark. *This is exactly the optimal-feedback form $\mathbf{v}^* = v_{\max}\nabla V/\|\nabla V\|$ of Theorem 2.1 with the intractable value V replaced by the tractable one-step surrogate J (and its shaping correction). Summing per-term gradients $-\nabla E = \sum_k(-\nabla E_k)$ realises the "force field": each named force is the negative gradient of one potential well, so their vector sum* is $-\nabla E$*.*

6 Diffusion as anticipation: the length scale

Why mollify J with a kernel of a *specific* width? Because the decision at $\mathbf{z}$ is exposed to the diffusive displacement of (1) before it can act again. Averaging a potential over that displacement is a heat semigroup.

Theorem 6.1 (Heat semigroup / Feynman–Kac [3, Ch. 4–5], [4, Ch. 2])**.** *Let $Z_\tau = \mathbf{z} + \sigma W_\tau$ be the (uncontrolled) diffusion started at $\mathbf{z}$ and let U be bounded measurable. Then $u(\tau, \mathbf{z}) := \mathbb{E}[U(Z_\tau)]$ solves the heat equation $\partial_\tau u = \frac{\sigma^2}{2}\Delta u$, $u(0,\cdot) = U$, and is given by convolution with the Gaussian heat kernel,*

$$\mathbb{E}[U(Z_\tau)] \;=\; \big(G_{\sigma^2\tau} * U\big)(\mathbf{z}), \qquad G_{s^2}(\mathbf{x}) = \tfrac{1}{2\pi s^2}\exp\big(-\|\mathbf{x}\|^2/2s^2\big).$$

Lemma 6.2 (Gaussian convolution: variances add)**.** *For $a, s > 0$ in $\mathbb{R}^2$, $G_{a^2} * G_{s^2} = G_{a^2+s^2}$.*

Proof. The Fourier transform of G_{a^2} is $\widehat{G_{a^2}}(\xi) = e^{-a^2\|\xi\|^2/2}$. Convolution becomes multiplication under $\widehat{\cdot}$, so $\widehat{G_{a^2} * G_{s^2}} = e^{-a^2\|\xi\|^2/2}e^{-s^2\|\xi\|^2/2} = e^{-(a^2+s^2)\|\xi\|^2/2} = \widehat{G_{a^2+s^2}}$. Injectivity of the Fourier transform gives the claim [5, Ch. 8]. □

Corollary 6.3 (Anticipation inflates the interaction radius)**.** *Model a danger/interaction zone of native radius r_{lim} by a Gaussian well $U(\mathbf{z}) = G_{r^2_{\text{lim}}}(\mathbf{z} - a)$ (up to scale). Its diffusive average over a horizon τ is, by Theorem 6.1 and Lemma 6.2,*

$$\mathbb{E}[U(Z_\tau)] \;=\; \big(G_{\sigma^2\tau} * G_{r^2_{\text{lim}}}\big)(\mathbf{z} - a) \;=\; G_{\ell^2}(\mathbf{z} - a), \qquad \boxed{\ell = \sqrt{r^2_{\text{lim}} + \sigma^2\tau}}.$$

Remark (Implementation and the small-time expansion)**.** *With a horizon of $\tau = \tau_{\text{steps}}\Delta t$, the inflated scale is $\ell = \sqrt{r^2_{\text{lim}} + \sigma^2\tau_{\text{steps}}\Delta t}$, i.e. the code's $\ell^2 = r^2_{\text{lim}} + (\sigma\sqrt{\Delta t}\,\sqrt{\tau_{\text{steps}}})^2$. To first order, Itô/Taylor gives $\mathbb{E}[U(Z_\tau)] = U(\mathbf{z}) + \frac{\sigma^2\tau}{2}\Delta U(\mathbf{z}) + o(\tau)$ [2], reproducing the $\frac{\sigma^2}{2}\Delta$ term of the HJB equation (3): the widening is the leading diffusive correction to the myopic value. Practically, the player begins avoiding Kaya and stronger rivals* before *contact, because diffusion may close the gap within the horizon.*

7 The crowd term as a mean field

Definition 7.1 (Empirical measure and crowd potential). *Let* $\mu_N = \frac{1}{N}\sum_{j=1}^{N} \delta_{\mathbf{z}_j}$ *be the empirical measure of player positions and, for a short-range repulsive kernel* $W \geq 0$,

$$E_{\text{crowd}}(\mathbf{z}) \;=\; (W * \mu_N)(\mathbf{z}) \;=\; \frac{1}{N}\sum_{j=1}^{N} W(\mathbf{z} - \mathbf{z}_j). \tag{8}$$

Theorem 7.2 (Convergence of empirical measures [8]). *If* $\mathbf{z}_1, \mathbf{z}_2, \ldots$ *are i.i.d. with law* μ *on a separable metric space, then almost surely* $\mu_N \to \mu$ *weakly as* $N \to \infty$*. Consequently, for bounded continuous* $W(\mathbf{z} - \cdot)$, $E_{\text{crowd}}(\mathbf{z}) = \int W(\mathbf{z} - y)\, d\mu_N(y) \to \int W(\mathbf{z} - y)\, d\mu(y)$ *a.s.*

Remark (Mean-field scaling and best response). *The* $1/N$ *normalisation is the mean-field scaling that keeps* $E_{\text{crowd}} = O(1)$ *as the field grows, so no single decision is dominated by raw crowd size; only the* local density $\int_{\|\mathbf{z}-y\| \lesssim r_{\text{lim}}} d\mu_N$ *matters, which is what* (8) *measures. Choosing* $\mathbf{v} = -\nabla E$ *against the population density* μ_N *is precisely a* best response to the mean field*; the fixed point in which every player best-responds to the induced density is the Nash concept of mean-field games [9, 10], whose* $N \to \infty$ *justification is propagation of chaos [11]. Players here are neither i.i.d. nor exchangeable at all times, so Theorem 7.2 is invoked as a modelling approximation, not an exact law; its role is to justify the functional form* (8) *and the* $1/N$ *weight. Behaviourally,* E_{crowd} *repels the player from clusters, where concurrent duels inflate the variance of* $\sum_j \Delta H$ *and invite being ganged.*

8 Gradient-flow stability and the camp

Theorem 8.1 (Lyapunov descent and LaSalle [12, 13]). *Along the (uncapped) gradient flow $\dot{\mathbf{z}} = -\nabla E(\mathbf{z})$ on the compact set $\bar{D}$ with $E \in C^1$, the energy is nonincreasing, $\frac{d}{dt}E(\mathbf{z}(t)) = -\|\nabla E(\mathbf{z}(t))\|^2 \leq 0$. Every trajectory converges to the largest invariant subset of the critical set $\{\nabla E = 0\}$; if the critical points are isolated, trajectories converge to a single critical point.*

Proof. $\frac{d}{dt}E(\mathbf{z}) = \langle \nabla E, \dot{\mathbf{z}} \rangle = -\|\nabla E\|^2 \leq 0$, so E is a Lyapunov function. $\bar{D}$ is compact and E continuous, hence sublevel sets are compact; LaSalle's invariance principle [13] gives convergence to the largest invariant set inside $\{\dot{E} = 0\} = \{\nabla E = 0\}$. □

Remark (Capped flow). *Under the projected flow (7), $\frac{d}{dt}E = \langle \nabla E, \mathbf{v} \rangle = -v_{\max}\|\nabla E\| \leq 0$ whenever $\nabla E \neq 0$, so descent and the LaSalle conclusion persist; the cap changes speed, not monotonicity.*

Proposition 8.2 (Stable camp radius). *Let the radial part of E_{home} be $\varphi(r) = \frac{1}{2}w\,(r - r^*)^2$ with $w > 0$, $r = \|\mathbf{z}\|$. Then $r = r^*$ is the unique critical radius and it is asymptotically stable for the radial flow $\dot{r} = -\varphi'(r)$.*

Proof. $\varphi'(r) = w(r - r^*)$ vanishes only at $r = r^*$, and $\varphi''(r^*) = w > 0$, so r^* is a strict local (indeed global) minimum of φ; by Theorem 8.1 applied to the scalar flow, $r(t) \to r^*$. □

Remark (Why the boundary). *The angular dynamics are governed by E_{crowd}, which spreads players apart along the ring $r = r^*$; the radial dynamics pin them near the rim. The choice r^* just inside the recovery band $\{r \geq 1 - r'\}$ is deliberate: area grows like r, so a boundary camp has fewer expected neighbours (hence, by Lemma 4.1, less negative fight flux) than the interior, while keeping the recovery region one step away. When H falls, r^* is shifted into the band and w is increased $\propto \beta$, committing the player to healing; this is a state-dependent deformation of the same convex well, preserving Proposition 8.2. The full E is nonconvex (each rival adds a well), so only* local *stability is claimed — consistent with a game whose global configuration is adversarial.*

9 The finite-horizon aggression schedule

Lemma 3.1 showed every fight has $\mathbb{E}[\Delta H \mid \text{fight}] \le 0$; a strictly health-greedy player (the surrogate $E = -J$ alone) therefore never initiates combat. Yet the terminal objective (2) rewards being *last*, which requires others to reach 0. The shaping term $-g_{\text{end}}\Phi_{\text{prey}}$ restores this, with

$$\Phi_{\text{prey}}(\mathbf{z}) = \sum_{j:\, s_{ij}>0} s_{ij}\, K_\ell(\mathbf{z} - \mathbf{z}_j)\, g_{\text{HP}}\, \text{iso}_j, \tag{9}$$

an attractor toward weak ($s_{ij} > 0$), isolated (iso_j small-crowd) prey, active only when the player is healthy (g_{HP}). The gate $g_{\text{end}} \in [0, 1]$ increases as the field empties.

Proposition 9.1 (Vanishing option value of waiting)**.** *Let $k(t)$ be the number of living players and suppose among the $k - 1$ opponents combat eliminations arrive at aggregate rate $\Lambda_{\text{opp}}(k)$ with $\Lambda_{\text{opp}}(2) = 0$ and Λ_{opp} increasing in k. Then the expected number of "free" opponent-deaths obtained by a passive player over a short horizon h is $\Lambda_{\text{opp}}(k)\, h + o(h)$, which is increasing in k and vanishes at $k = 2$.*

Proof. Free deaths accrue as a counting process with intensity $\Lambda_{\text{opp}}(k)$; its expected increment over $[0, h]$ is $\Lambda_{\text{opp}}(k)h + o(h)$ [6]. Monotonicity in k and the boundary value $\Lambda_{\text{opp}}(2) = 0$ are the hypotheses. $\square$

Remark (Reading the schedule)**.** *Proposition 9.1 formalises the trade-off: when many rivals remain, the field culls itself for free, so paying HP to attack is wasteful and $g_{\text{end}} \approx 0$ (patient camping is near-optimal). As $k \downarrow 2$ the free-death rate $\to 0$; the only path to the terminal reward is to spend a little HP closing on the weakest survivor, so $g_{\text{end}} \to 1$. The schedule $g_{\text{end}}(k)$ is thus a monotone interpolation, the myopically-invisible part of the value function reinstated by rollout reasoning [7, Ch. 6]. Simulation over 10^2 tournaments corroborates the qualitative claim: a small* persistent *predation baseline (culling weak isolated rivals) outperforms strict pacifism, because it shrinks the survivor pool faster than it costs HP, while heavy early aggression underperforms.*

10 No policy guarantees a win

Proposition 10.1 (Irreducible elimination risk)**.** *Fix any horizon $T > 0$ and any admissible feedback controls for all players ($\|\mathbf{v}_i\| \le v_{\max} < \infty$). While at least two players remain, each player is eliminated with strictly positive probability by time T; in particular no admissible policy attains win probability* 1*.*

Proof. The reflected diffusion (1) on the compact domain $\bar{D}$ has, for each $t > 0$, a transition density that is bounded below by a strictly positive Gaussian-type kernel, uniformly on $\bar{D}$ (Aronson's lower bound for uniformly parabolic operators [14]). A bounded drift $\mathbf{v}$, $\|\mathbf{v}\| \le v_{\max}$, only reweights this density by a factor bounded away from 0 (Girsanov [3]), so it cannot annihilate any positive-probability event. Hence with positive probability two given players occupy the same $r_{\lim}$-neighbourhood on a time interval of positive length. Conditioned on that, the fight clock (rate $\lambda > 0$) fires at least once with probability $1 - e^{-\lambda\cdot(\mathrm{dwell})} > 0$ [6], and by Lemma 3.1 the focal player loses that exchange with probability $H_j/(H_i + H_j) \in (0, 1)$. Chaining these positive-probability events shows $\mathbb{P}[\Delta H_i < 0 \text{ on } [0, T]] > 0$; iterating over the finite HP budget gives $\mathbb{P}[i \text{ eliminated by } T] > 0$ for T large enough. Therefore $\mathbb{P}[i \text{ wins}] \le 1 - \mathbb{P}[i \text{ eliminated}] < 1$. □

Remark. *Proposition 10.1 is the precise sense in which "guarantee a win" is unattainable: the uncontrolled diffusion has full support on a compact arena, so no amount of steering removes the possibility of a losing encounter. The design goal is therefore correctly stated as* maximising win probability*, which is what the surrogate $-E$ and its shaping term target, and what the simulations measure (a* 6–7× *uplift over the uniform baseline* $1/N$*).*

11 Intuition: reading the energy as behaviour

Each mathematical term corresponds to a legible instinct; the policy is their vector sum $-\nabla E$, capped at $v_{\max}$.

- **Fear Kaya (repulsion).** Kaya is a moving Gaussian danger well (Lemma 4.1); its force pushes radially away, amplified by a marginal-value-of-HP factor $\varphi(H)$ that grows as $H \to 0$, because near elimination the survival value function is steep — one HP is worth more when you have few.

- **Duel selectively (signed attraction).** The coefficient s_{ij} of Lemma 3.1 makes healthier-than-me rivals repel and weaker-isolated ones (mildly) attract: fight only the fights the win law makes favourable, and flee the rest.

- **Avoid crowds (mean field).** E_{crowd} (§7) keeps the player out of dense clusters, where simultaneous duels create ruinous HP variance.

- **Heal when wounded (conditional attraction).** Butter attracts only while $H < 50$ and in proportion to the healable deficit $(50 - H)_+$; below 25 the radial term commits the player into the boundary recovery band with strength $\propto \beta$.

- **Camp the rim (stable equilibrium).** The radial well (Proposition 8.2) parks the player just inside the boundary — fewest neighbours, healing adjacent — while crowd repulsion spreads campers around the ring.

- **Anticipate the noise (length scale).** Every interaction radius is inflated to $\ell = \sqrt{r_{\text{lim}}^2 + \sigma^2 \tau}$ (Corollary 6.3), so the player reacts to where diffusion *could* put it, not only where it is.

- **Finish late (finite-horizon shaping).** While many remain, wait and let the field thin itself (Proposition 9.1); as the count falls to two, turn aggressor and close on the weakest survivor.

The composite is a patient, boundary-hugging survivor that dodges the cat and the strong, tops up its health when low, quietly removes weak stragglers, and turns killer only when the arithmetic of being "last" finally demands it.

Summary of results

Result	Statement	Main tool
Thm. 2.1 (HJB)	control enters via $\mathbf{v} \cdot \nabla V$; optimum saturates cap along a gradient	DPP + verification [1]
Lem. 3.1 (fights)	$\mathbb{E}[\Delta H_i \mid \text{fight}] = -\frac{H_j}{H_i+H_j}$, advantage s_{ij}	elementary expectation
Lem. 4.1 (flux)	health drift $J(\mathbf{z})$ from pooled Poisson rates	Poisson superposition [6]
Lem. 5.1 (ascent)	$\arg\max_{\\|\mathbf{v}\\| \le c} \langle g, \mathbf{v} \rangle = c\, g/\\|g\\|$	Cauchy–Schwarz
Thm. 6.1, Cor. 6.3	$\ell = \sqrt{r_{\text{lim}}^2 + \sigma^2 \tau}$ (anticipation)	heat semigroup [3], Gaussian convolution
Thm. 7.2 (mean field)	$\mu_N \to \mu$; $1/N$ crowd scaling	LLN for empirical measures [8]
Thm. 8.1, Prop. 8.2	gradient flow descends E; stable camp radius	Lyapunov / LaSalle [12, 13]
Prop. 9.1 (schedule)	free-death rate $\to 0$ as $k \to 2$	counting-process intensity
Prop. 10.1	no policy wins with probability 1	Aronson bound [14], Girsanov

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